CAPSTONE PROJECT

NUTRITION AGENT

Presented By:
Harshita Gupta – Dr. Akhilesh Das Gupta Institute of Professional Studies

edu

OUTLINE

Problem Statement

Proposed System/Solution

System Development Approach (Technology Used)

Algorithm & Deployment

Result (Output Image)

Conclusion

Future Scope

References

ROBLEM STATEMENT

n era where health awareness is growing, individuals increasingly
k personalized nutrition guidance. However, most existing tools
vide generic diet plans, lack real-time adaptability, and fail to
sider a person's holistic lifestyle, cultural preferences, allergies, and
lving health conditions. Furthermore, dieticians and nutritionists fac
tations in scaling personalized consultations due to time and
ource constraints.

ROPOSED SOLUTION

Glow:

utriGlow is an AI-powered nutrition and wellness chatbot designed to offer personalized health advice thro
mple and intuitive interface. Hosted on IBM Cloud and developed using IBM Watsonx Assistant, NutriGlow
ers:

hoose their health goals (e.g., weight loss, weight gain, better nutrition).

et recommended yoga asanas tailored to their chosen goal.

cess personalized healthy diet plans based on their preferences and objectives.

plore additional resources like notes and syllabus (if expanded to academic nutrition programs).

egrating natural language processing and intelligent flow design, NutriGlow provides 24/7 assistance for
prove their health in a guided, personalized, and interactive way.

YSTEM APPROACH

em Design

rchitecture:

- **Frontend:** HTML, Tailwind CSS (for website UI)
- **Backend:** IBM Watsonx Assistant (chatbot logic)
- **Hosting:** IBM Cloud

Modules:

- Goal Selection Module
- Yoga & Exercise Recommendation Engine
- Diet Plan Generator
- User Interaction Interface (chatbot)

low:

- Greet user
- Ask for user's goal
- Recommend yoga asanas based on goal
- Provide customized diet plan

LGORITHM & DEPLOYMENT

loyment Steps:

Create the Chatbot Logic in IBM Watson Assistant:

- Create Intents (e.g., #weight_loss, #diet_plan)
- Define Entities (e.g., @goal, @food_type)
- Build Dialog nodes to manage conversation flow.

Design the Website UI:

- Use HTML & Tailwind CSS to build a clean, responsive frontend.
- Embed the chatbot using Watson Assistant web chat integration script.

Host the Website:

- Use IBM Cloud to create a static web app or use IBM Cloud Object Storage with public access.

LGORITHM & DEPLOYMENT

Link the Frontend with the Chatbot:

- Paste Watson Web Chat script into the HTML <body>.
- Ensure proper assistant ID, integration ID, and region are used.

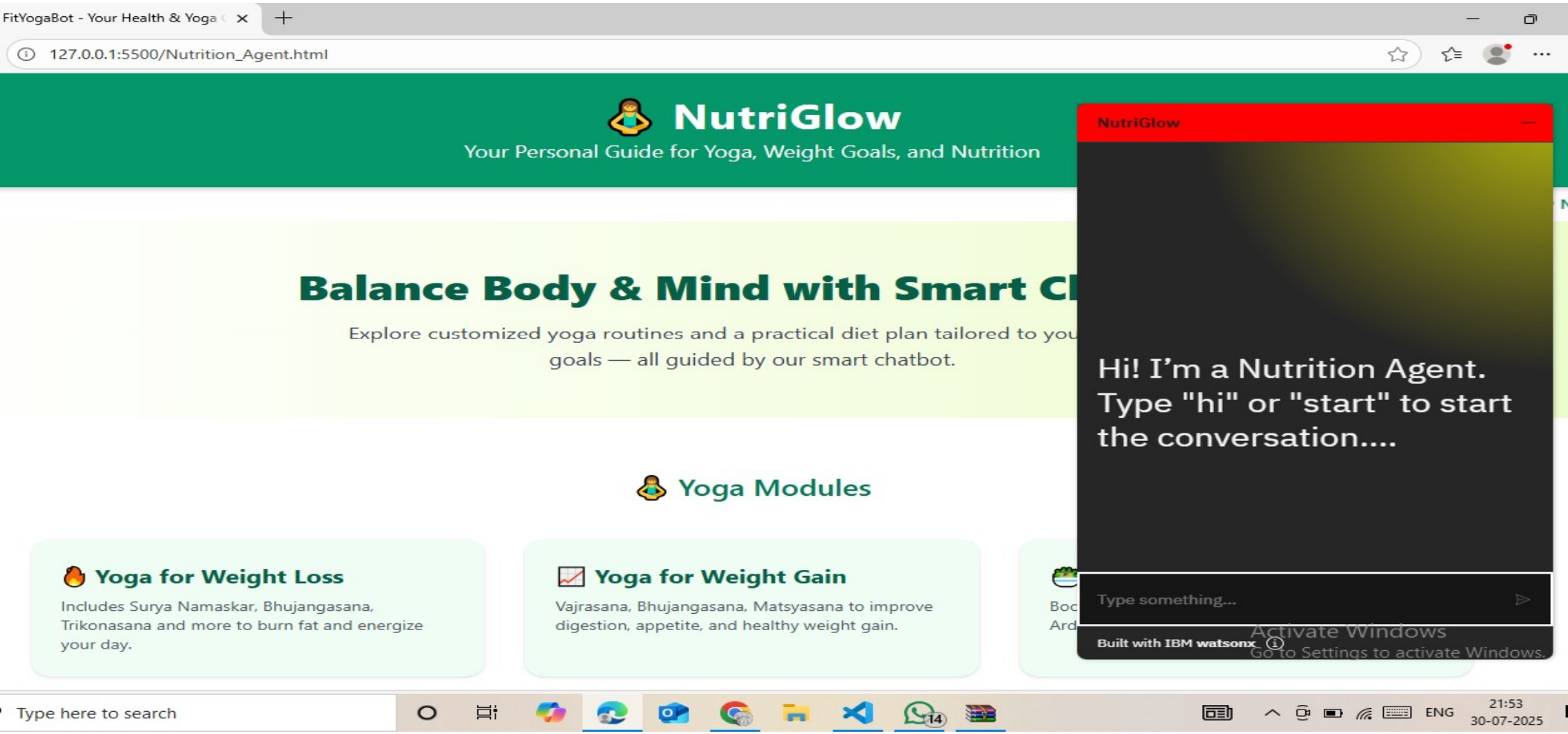
Test Deployment:

- Interact with the chatbot on the hosted site.
- Verify that all paths (weight loss/gain, diet, resources) work properly.

Go Live & Share:

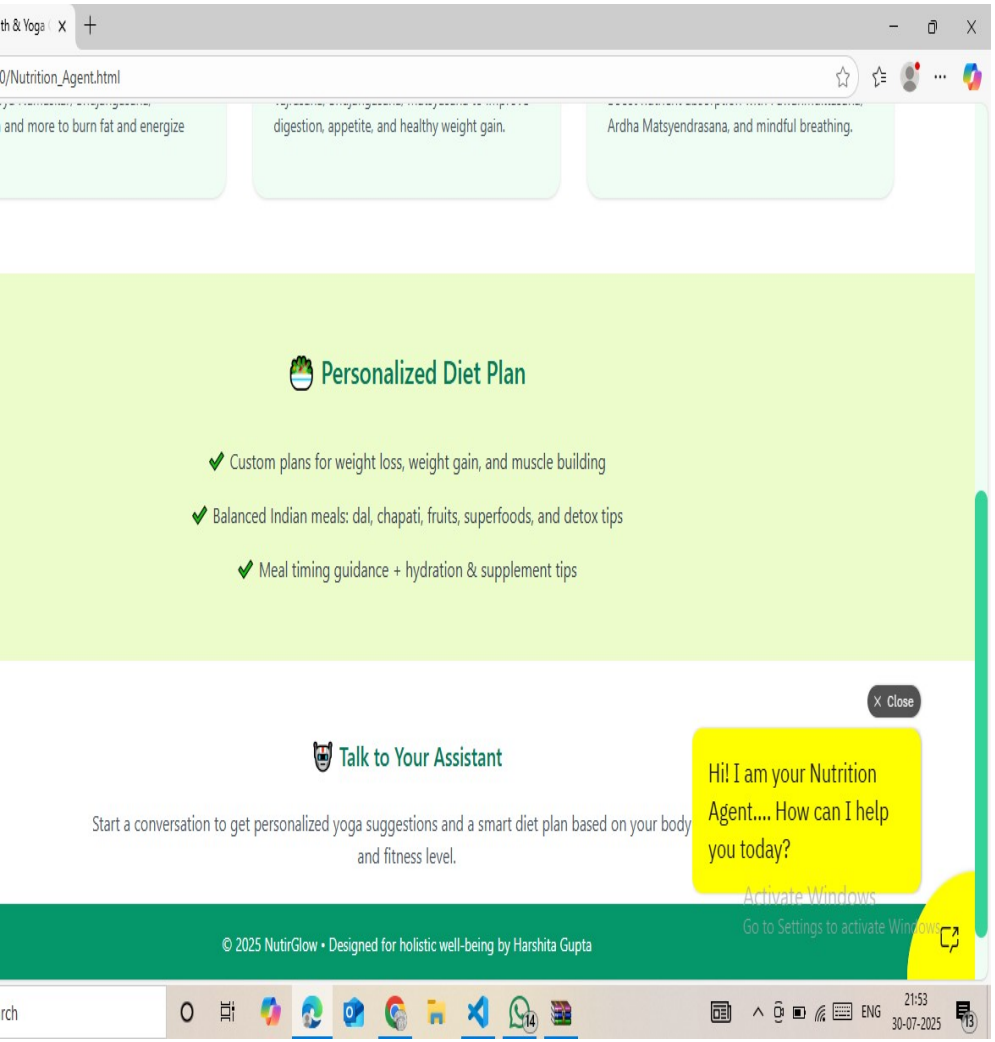
- Share the IBM Cloud public URL with users.

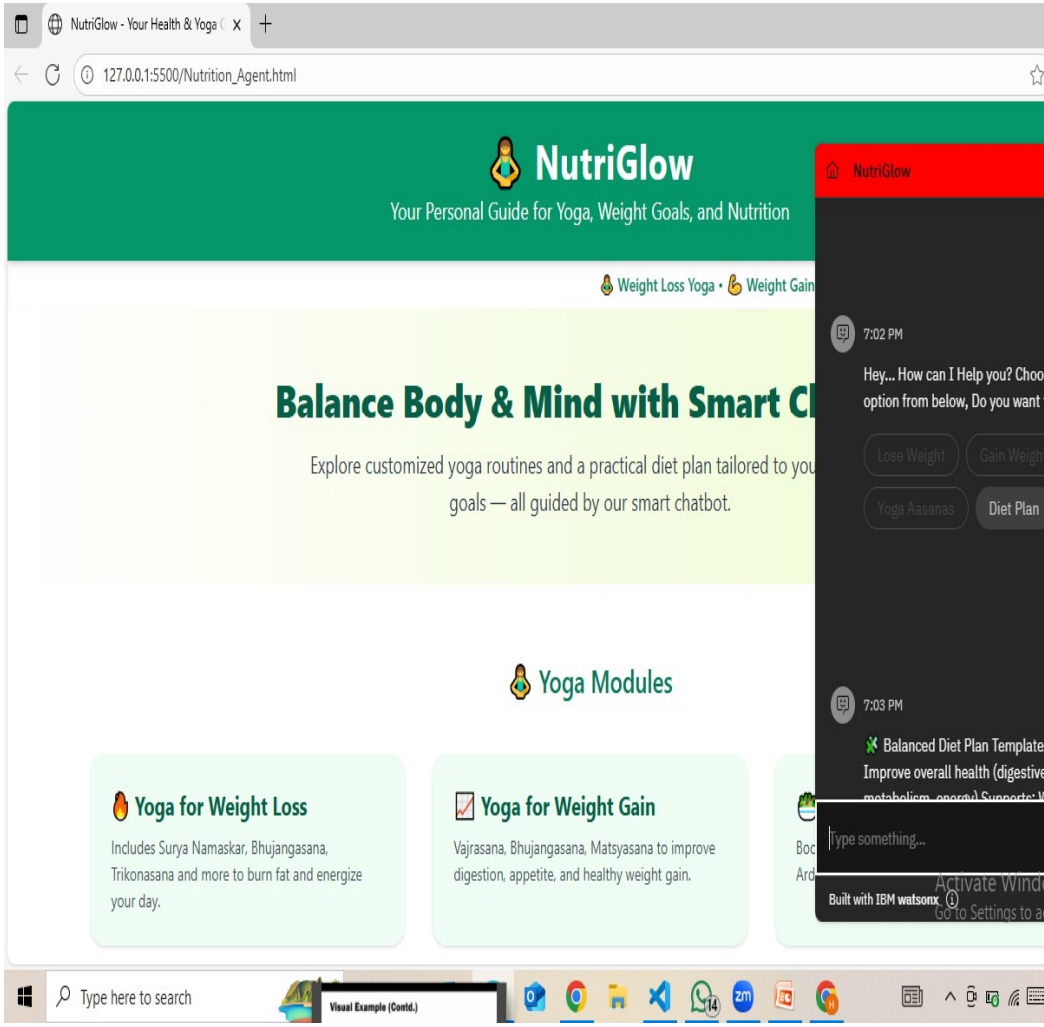
ESULT


FitYogaBot - Your Health & Yoga
127.0.0.1:5500/Nutrition_Agent.html
NutriGlow
Your Personal Guide for Yoga, Weight Goals, and Nutrition
Balance Body & Mind with Smart Cl
Explore customized yoga routines and a practical diet plan tailored to you
goals — all guided by our smart chatbot.
Yoga Modules
Yoga for Weight Loss
Includes Surya Namaskar, Bhujangasana, Trikonasana and more to burn fat and energize your day.
Yoga for Weight Gain
Vajrasana, Bhujangasana, Matsyasana to improve digestion, appetite, and healthy weight gain.
NutriGlow
Hi! I'm a Nutrition Agent. Type "hi" or "start" to start the conversation....
Type something...
Built with IBM watsonx
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
21:53
30-07-2025

edu

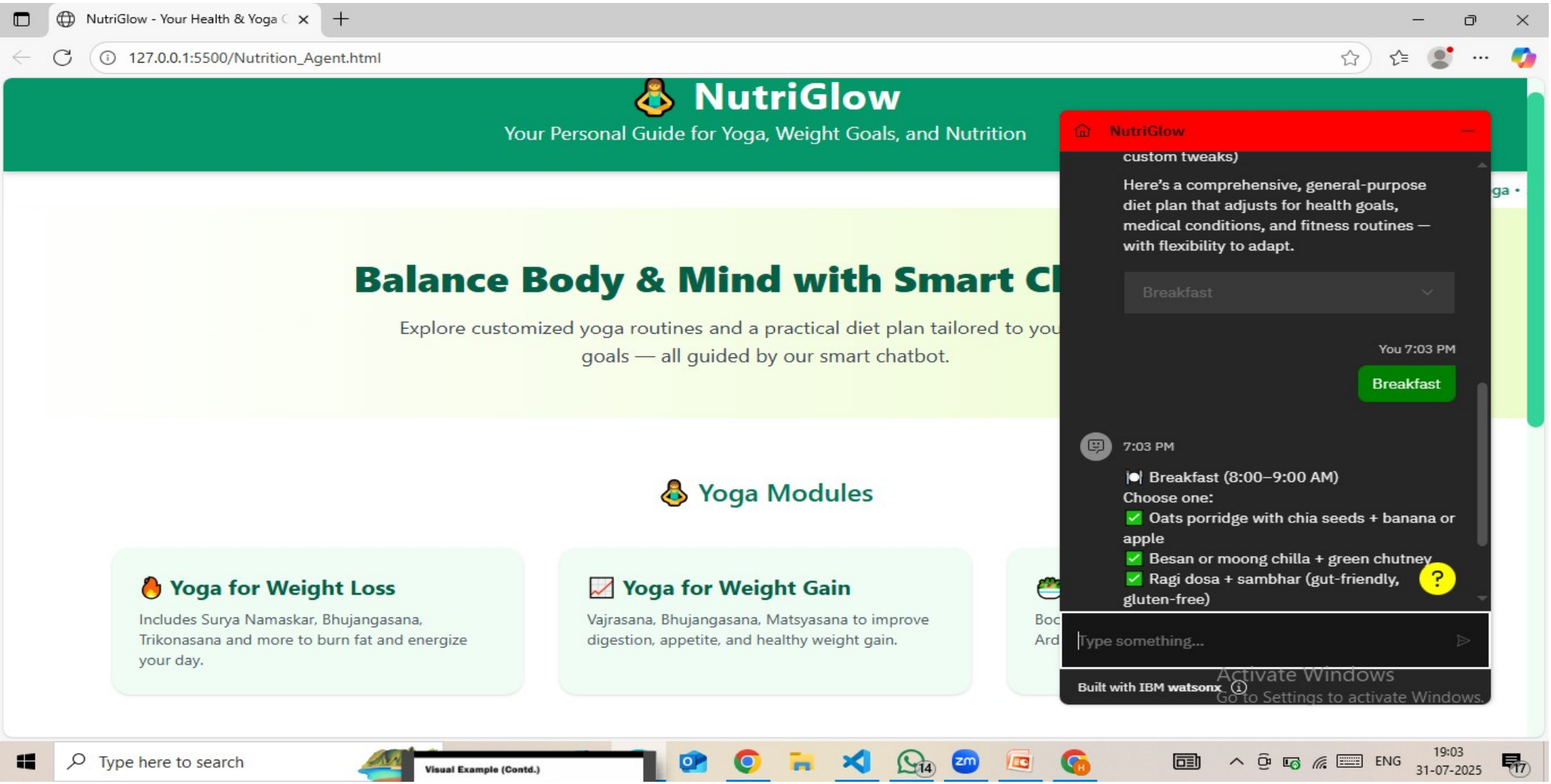
ESULT


th & Yoga
0/Nutrition_Agent.html
and more to burn fat and energize
digestion, appetite, and healthy weight gain.
Ardha Matsyendrasana, and mindful breathing.
Personalized Diet Plan
Custom plans for weight loss, weight gain, and muscle building
Balanced Indian meals: dal, chapati, fruits, superfoods, and detox tips
Meal timing guidance + hydration & supplement tips
X Close
Talk to Your Assistant
Start a conversation to get personalized yoga suggestions and a smart diet plan based on your body and fitness level.
Hi! I am your Nutrition Agent.... How can I help you today?
© 2025 NutirGlow • Designed for holistic well-being by Harshita Gupta
ENG
21:53
30-07-2025


NutriGlow - Your Health & Yoga
127.0.0.1:5500/Nutrition_Agent.html
NutriGlow
Your Personal Guide for Yoga, Weight Goals, and Nutrition
Weight Loss Yoga • Weight Gain
Balance Body & Mind with Smart C
Explore customized yoga routines and a practical diet plan tailored to you
goals — all guided by our smart chatbot.
Yoga Modules
Yoga for Weight Loss
Includes Surya Namaskar, Bhujangasana, Trikonasana and more to burn fat and energize your day.
Yoga for Weight Gain
Vajrasana, Bhujangasana, Matsyasana to improve digestion, appetite, and healthy weight gain.
NutriGlow
7:02 PM
Hey... How can I Help you? Choo
option from below, Do you want
Lose Weight
Yoga Aasanas
Diet Plan
7:03 PM
Balanced Diet Plan Template
Improve overall health (digestive
Type something...
Built with IBM watsonx
Type here to search
Visual Example (Contd.)

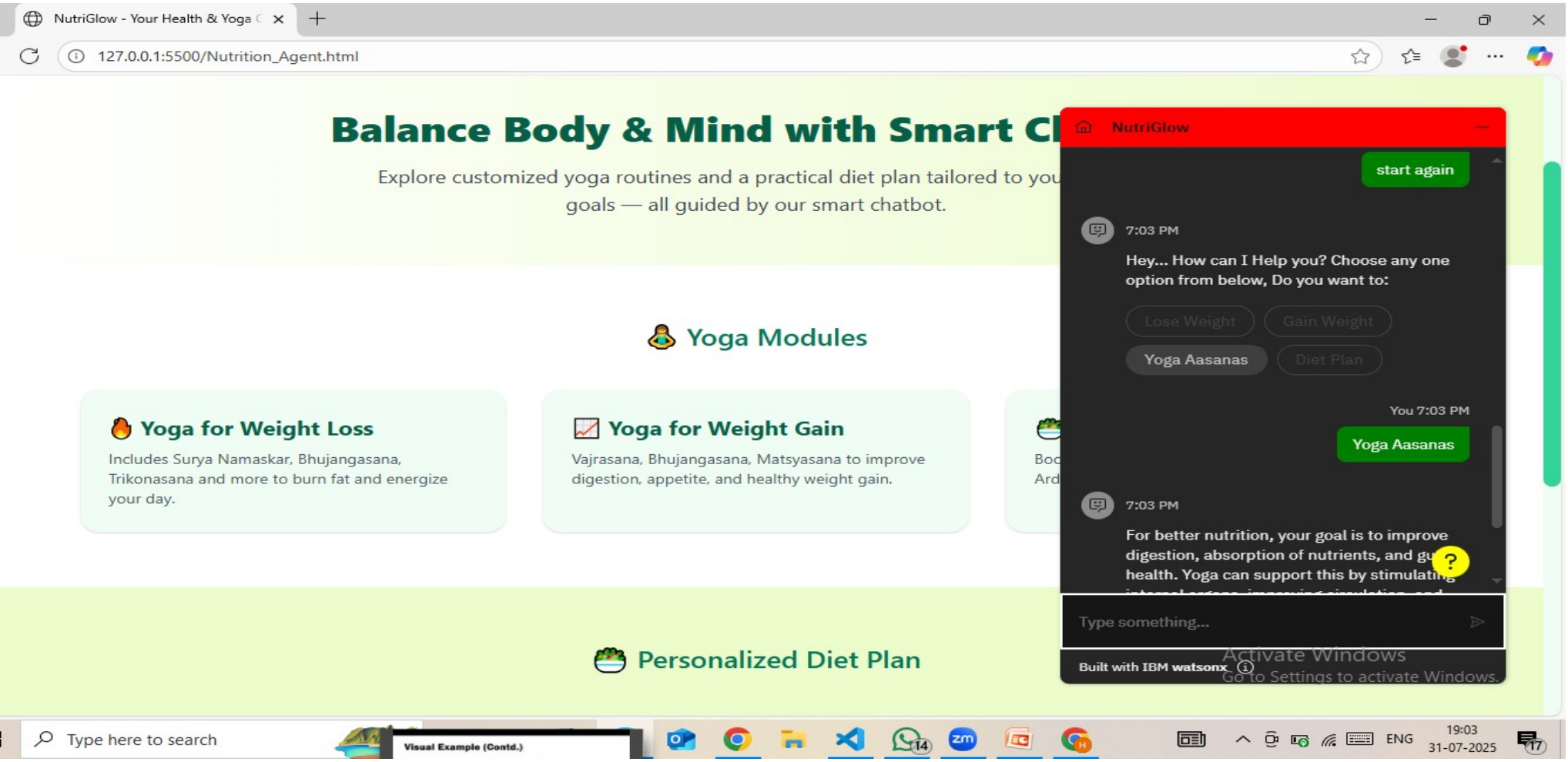
edu

ESULT


NutriGlow - Your Health & Yoga
127.0.0.1:5500/Nutrition_Agent.html
NutriGlow
Your Personal Guide for Yoga, Weight Goals, and Nutrition
Balance Body & Mind with Smart Cl
Explore customized yoga routines and a practical diet plan tailored to you
goals — all guided by our smart chatbot.
Yoga Modules
Yoga for Weight Loss
Includes Surya Namaskar, Bhujangasana, Trikonasana and more to burn fat and energize your day.
Yoga for Weight Gain
Vajrasana, Bhujangasana, Matsyasana to improve digestion, appetite, and healthy weight gain.
NutriGlow
custom tweaks)
Here's a comprehensive, general-purpose diet plan that adjusts for health goals, medical conditions, and fitness routines — with flexibility to adapt.
Breakfast
You 7:03 PM
Breakfast
7:03 PM
Breakfast (8:00–9:00 AM)
Choose one:
Oats porridge with chia seeds + banana or apple
Besan or moong chilla + green chutney
Ragi dosa + sambhar (gut-friendly, gluten-free)
Type something...
Built with IBM watsonx
Activate Windows
Go to Settings to activate Windows.
Type here to search
Visual Example (Contd.)
ENG
19:03
31-07-2025

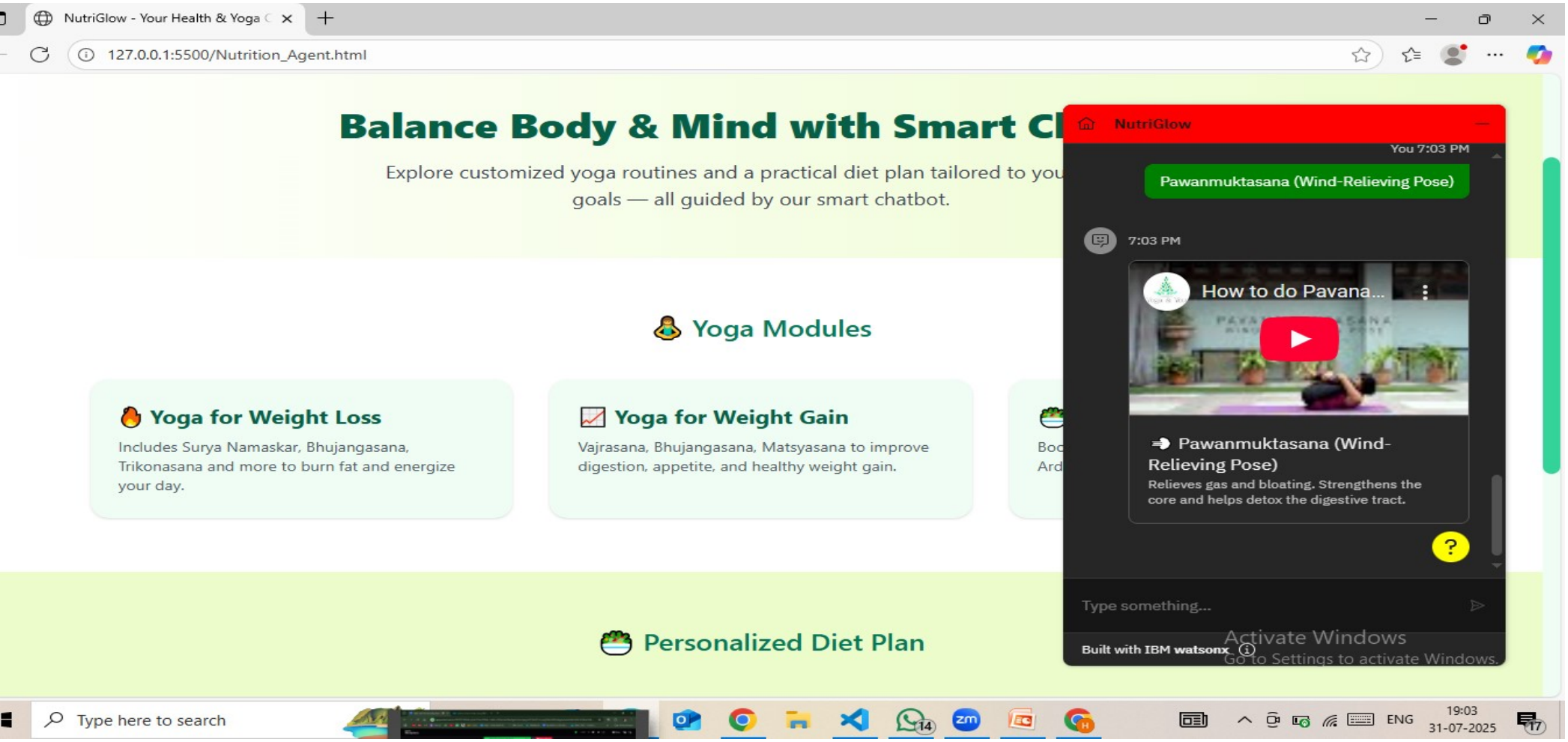
edu

ESULT


NutriGlow - Your Health & Yoga
127.0.0.1:5500/Nutrition_Agent.html
Balance Body & Mind with Smart Cl
Explore customized yoga routines and a practical diet plan tailored to you
goals — all guided by our smart chatbot.
Yoga Modules
Yoga for Weight Loss
Includes Surya Namaskar, Bhujangasana, Trikonasana and more to burn fat and energize your day.
Yoga for Weight Gain
Vajrasana, Bhujangasana, Matsyasana to improve digestion, appetite, and healthy weight gain.
Personalized Diet Plan
NutriGlow
start again
7:03 PM
Hey... How can I Help you? Choose any one option from below, Do you want to:
Lose Weight
Gain Weight
Yoga Aasanas
Diet Plan
You 7:03 PM
Yoga Aasanas
7:03 PM
For better nutrition, your goal is to improve digestion, absorption of nutrients, and gu
health. Yoga can support this by stimulati
Type something...
Built with IBM watsonx
Activate Windows
Go to Settings to activate Windows.
Type here to search
Visual Example (Contd.)
ENG
19:03
31-07-2025

edu

ESULT


NutriGlow - Your Health & Yoga
127.0.0.1:5500/Nutrition_Agent.html
Balance Body & Mind with Smart Cl
Explore customized yoga routines and a practical diet plan tailored to you
goals — all guided by our smart chatbot.
Yoga Modules
Yoga for Weight Loss
Includes Surya Namaskar, Bhujangasana, Trikonasana and more to burn fat and energize your day.
Yoga for Weight Gain
Vajrasana, Bhujangasana, Matsyasana to improve digestion, appetite, and healthy weight gain.
Personalized Diet Plan
NutriGlow
You 7:03 PM
Pawanmuktasana (Wind-Relieving Pose)
7:03 PM
How to do Pavana...
Pawanmuktasana (Wind-Relieving Pose)
Relieves gas and bloating. Strengthens the core and helps detox the digestive tract.
Type something...
Built with IBM watsonx
Activate Windows
Go to Settings to activate Windows.
Type here to search
ENG
19:03
31-07-2025

edu

ONCLUSION

NutriGlow successfully demonstrates how AI-powered chatbots can simplify and personalize hea
guidance for users. By combining yoga recommendations, nutritional advice, and customized di
plans in a user-friendly interface, the system addresses the growing need for accessible and
holistic wellness support. Through IBM Watsonx Assistant and cloud deployment, NutriGlow offe
a smart, scalable, and interactive solution that promotes healthier lifestyles without the need fo
expensive consultations. This project not only enhances user engagement in personal health bu
also showcases the potential of conversational AI in the field of digital wellness.

UTURE SCOPE

Jser Progress Tracking – Monitor diet, weight, and yoga activity over time.

Voice & Multilingual Support – Add voice interaction and regional languages.

Smart Device Integration – Connect with fitness bands for real-time health data.

Login System – Allow users to save preferences and track history.

EFERENCES

BM Watson Assistant Documentation

BM Cloud Hosting Guide

ailwind CSS Documentation

WHO – Healthy Diet Factsheet

M CERTIFICATIONS

In recognition of the commitment to achieve professional excellence

Harshita Gupta

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Jul 16, 2025
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/8dc74fcb-d0ab-484d-972d-f900e547b0d3

IBM

Activate Windows
Go to Settings to activate

M CERTIFICATIONS

In recognition of the commitment to achieve professional excellence

Harshita Gupta

Has successfully satisfied the requirements for:

Journey to Cloud: Envisioning Your Solution

Issued on: Jul 16, 2025
Issued by: IBM SkillsBuild

Verify: https://www.credly.com/badges/c27aa89e-6586-4704-9b06-3d57dc21340e

IBM

Activate Windows
Go to Settings to activate

M CERTIFICATIONS

demia_Problem Statement | NutriGlow Problem Statement | RAG Lab.pdf

C:/Users/Dell/Downloads/RAG%20Lab.pdf

Draw | Ask Copilot | 1 of 1

7/23/25, 6:38 PM Completion Certificate | SkillsBuild

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

Harshita Gupta

for the completion of

Lab: Retrieval Augmented Generation with LangChain

(ALM-COURSE_3824998)

According to the Adobe Learning Manager system of record

Completion date: 23 Jul 2025 (GMT) **Learning hours:** 20 mins

Activate Windows
Go to Settings to activate

here to search ENG

THANK YOU